TITANIC SURVIVAL ANALYSIS USING TABLEAU

Course: Interactive Data Visualization (IDV)

Team Members

- Rudra Pandit (ID – 202518040)
- Vrushil Panwala (ID – 202518010)
- Vishwa Patel (ID – 202518016)
- Keyur Pandya (ID – 202518019)
- Ram Prahasith (ID – 202518009)

Instructor: Dr. Bhaskar Chaudhury

Date: 4th April 2026

PROJECT GOALS AND NARRATIVE

- The goal of this project is to analyze survival patterns of passengers aboard the Titanic using data visualization techniques. The project aims to uncover key factors influencing survival such as gender, passenger class, age, and family size.

- The narrative of this project focuses on understanding how different demographic and social characteristics impacted survival during the Titanic disaster. Through visual storytelling, the project highlights inequalities and patterns in survival outcomes.

DATASET DESCRIPTION

It is about the Titanic ship that sank in 1912 after hitting an iceberg. The dataset has information about the passengers, including their age, gender, ticket class, family members on the ship, fare paid, cabin details, port of embarkation, and whether they survived or not. This dataset is used to study which factors affected the chances of survival.

Variables include:

- PassengerId: Unique identifier
- Pclass: Passenger class (1st, 2nd, 3rd)
- Sex: Gender
- Age: Age of passenger
- SibSp: Number of siblings/spouses
- Parch: Number of parents/children
- Fare: Ticket price
- Survived: Survival status

Derived Variables:

- Age Group (Child, Teen, Adult, Senior)
- Family Size = SibSp + Parch
- Family Category (Alone, Small Family, Large Family)

These derived features improved analytical depth.

PROBLEM STATEMENT

- The primary objective of this project is to identify and analyze the key factors that influenced passenger survival during the Titanic disaster.

- The challenge lies in understanding how different attributes such as gender, passenger class, age, and family size affected survival probability. This analysis aims to explore patterns and relationships between these variables using data visualization techniques.

- The goal is to derive meaningful insights that explain survival trends and highlight the most influential factors.

- Key questions addressed in this project include:
  - Did gender have a significant impact on survival?
  - How did passenger class affect the chances of survival?
  - What was the impact of age on survival probability?
  - Did family size influence survival chances?

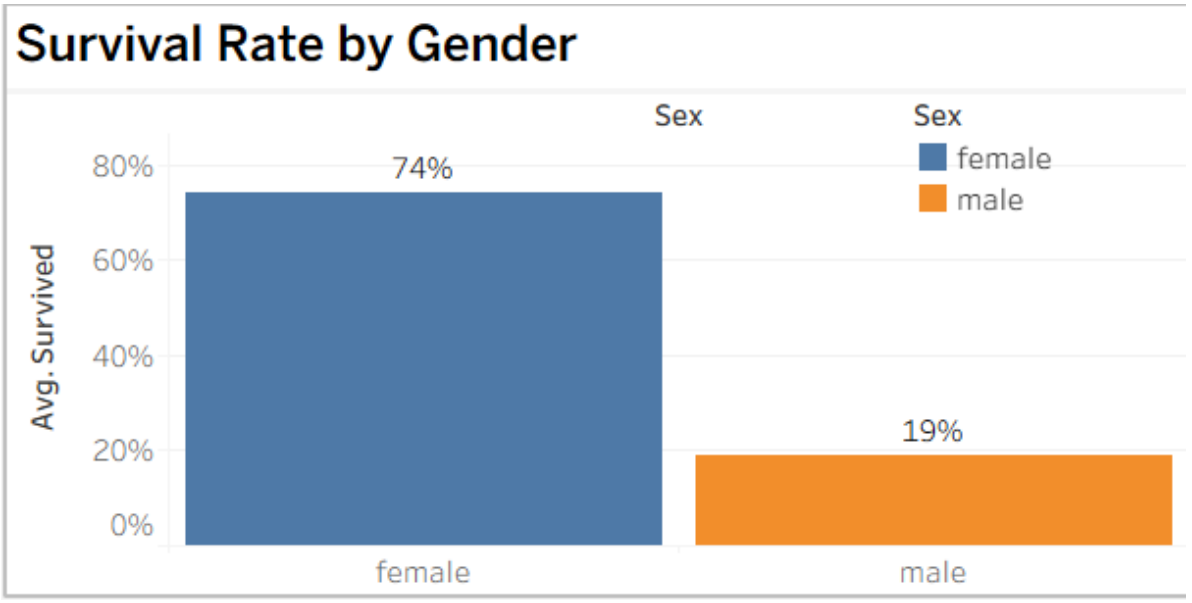
DATA PREPROCESSING AND TECHNIQUES

- The following steps were performed:
  - Data cleaning and handling missing values
  - Feature engineering (Age Group, Family Size, Family Category)
  - Data aggregation using AVG(Survived)
  - Categorization of variables

- Skills demonstrated:
  - Data transformation
  - Feature engineering
  - Tableau visualization
  - Dashboard design
  - Interactive filtering

➢ *Did gender the affect survival?*

GENDER VS SURVIVAL



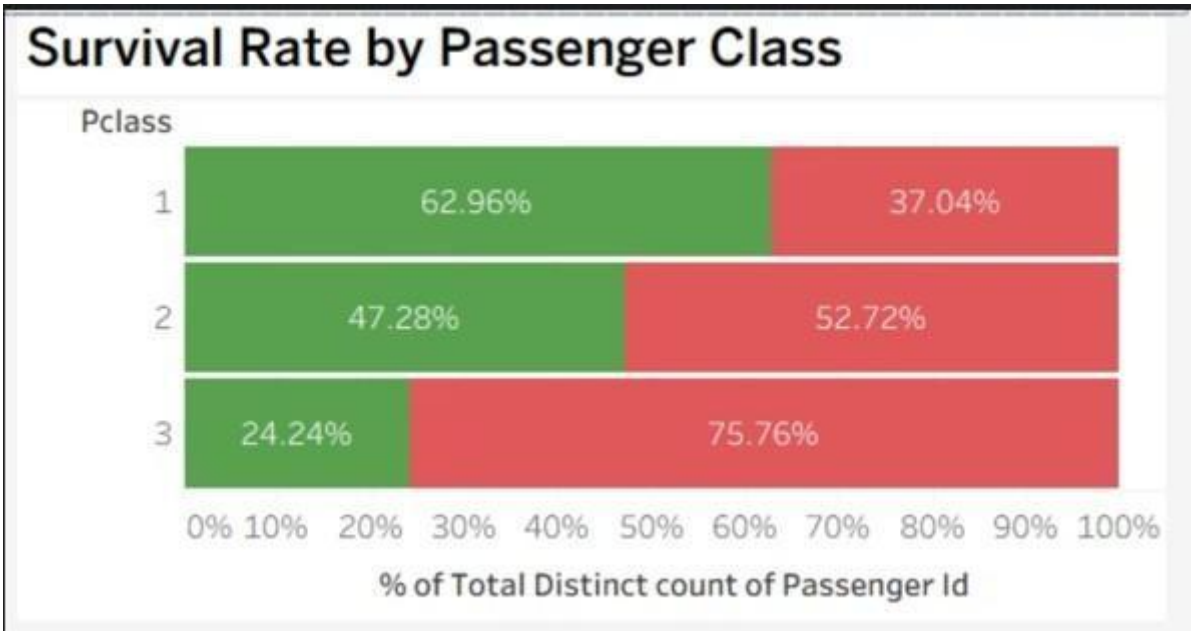
This visualization shows survival rates based on gender.

Observation:
Female passengers had a significantly higher survival rate (~74%) compared to males (~19%).

Rationale:
A bar chart was used for clear comparison between categories

- ***Did ticket class (1st, 2nd or 3rd class) affect survival?***

PASSENGER CLASS VS SURVIVAL

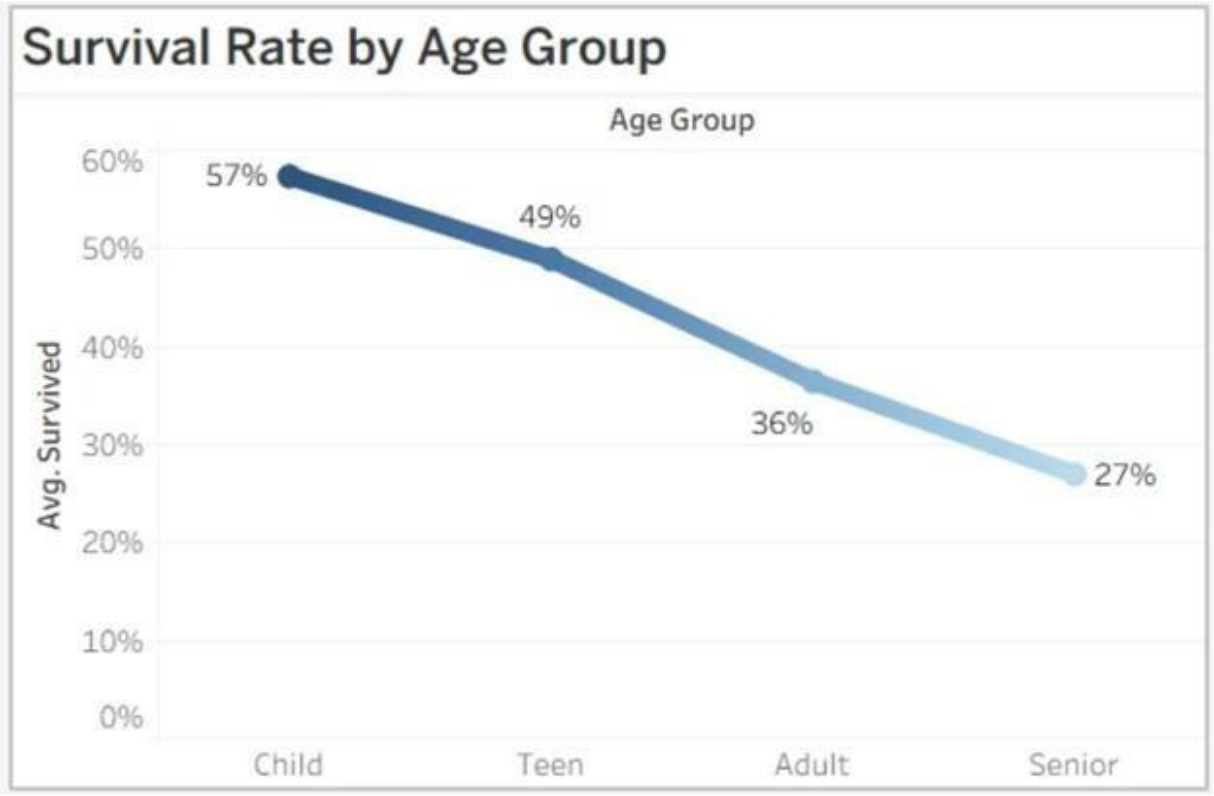


Observation:
First-class passengers had the highest survival (~63%), while third-class had the lowest (~24%).

Rationale:
Horizontal bar chart improves readability for class comparison.

➢ *Did age affect whether a passenger survived or not?*

AGE GROUP VS SURVIVAL

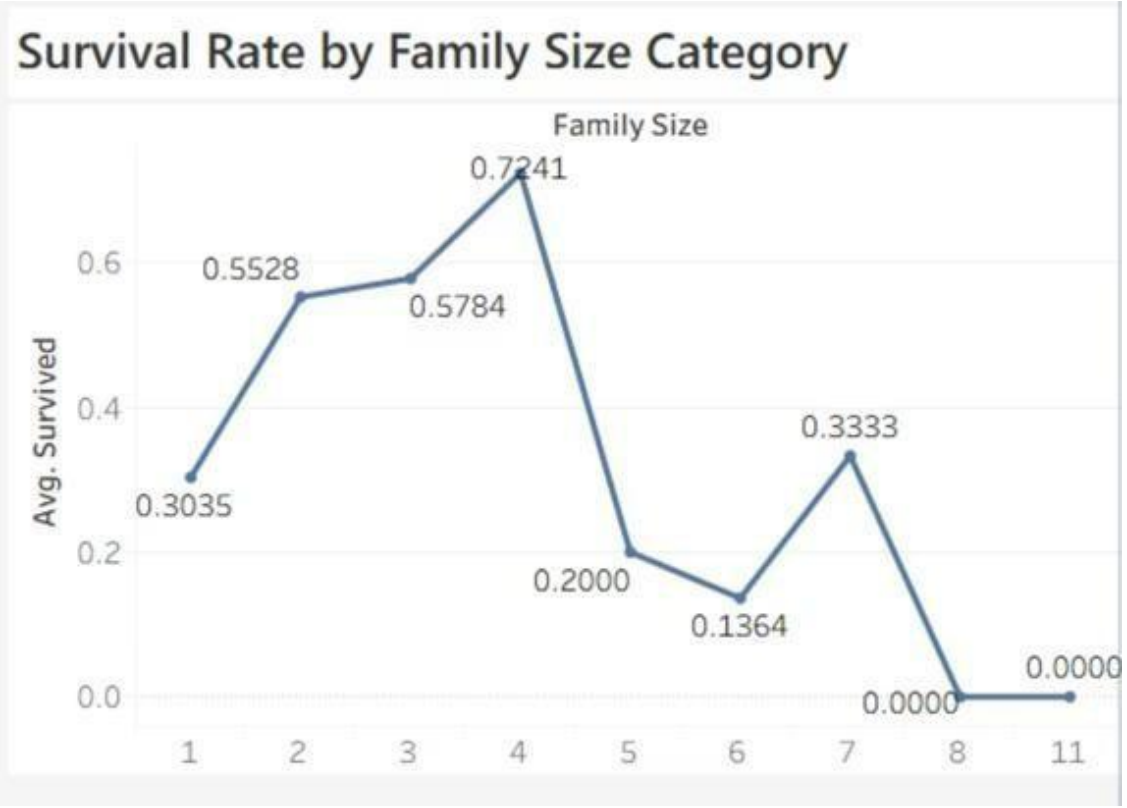


Observation:
Survival decreases with age. Children had the highest survival rate (~57%).

Rationale:
Line chart was used to show trend across age groups.

- ***How does the number of family members affect a person's chances of survival?***

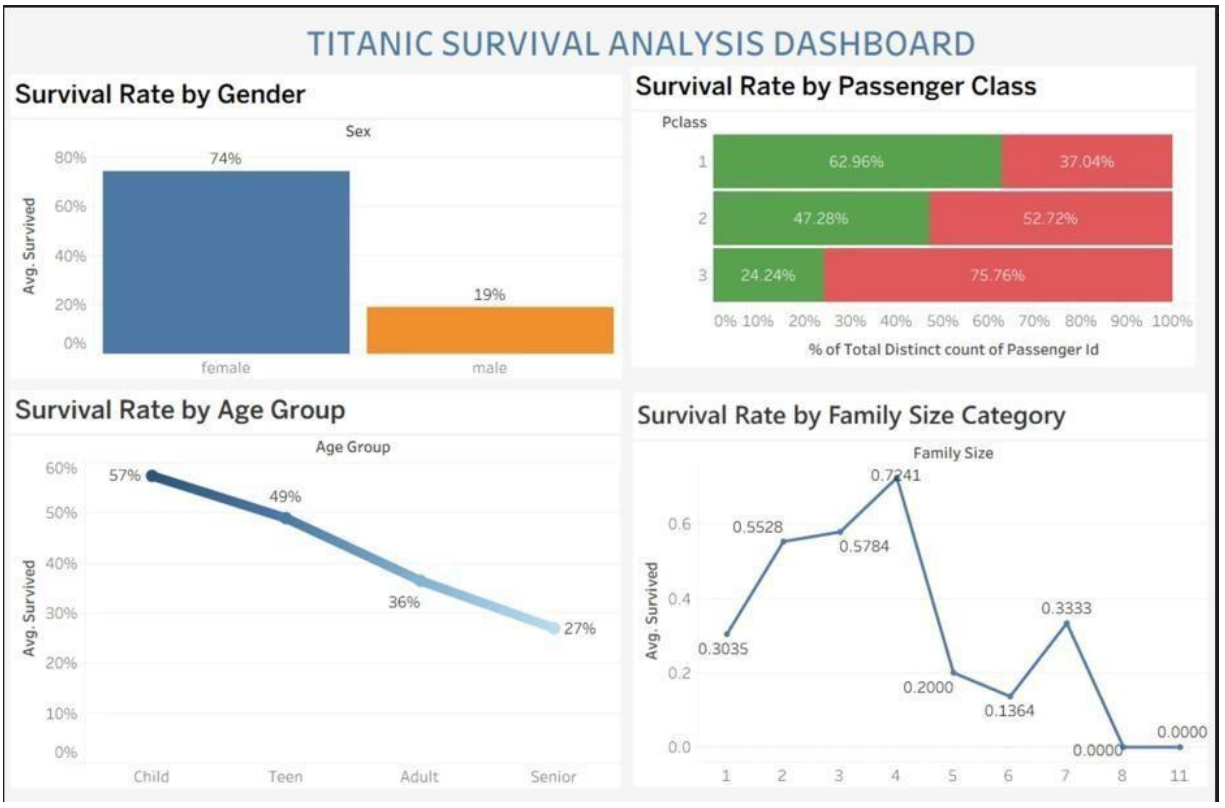
FAMILY SIZE VS SURVIVAL



Observation:
Small families had the highest survival (~58%), while large families had the lowest (~16%).

Rationale:
Bar chart effectively compares categorical groups.

TITANIC SURVIVAL ANALYSIS DASHBOARD



DASHBOARD DESCRIPTION

- The dashboard provides an overview of survival patterns in the Titanic dataset across gender, passenger class, age group, and family size.

- It combines multiple visualizations to allow comparison across different factors influencing survival. Interactive filters enable users to explore the data dynamically and gain deeper insights.

- Overall, the dashboard presents a clear and concise view of key survival trends.

VISUALIZATION TECHNIQUES AND DESIGN CHOICES

VISUALIZATION TECHNIQUES:

- Bar charts used for categorical comparison
- Line chart used for trend analysis
- Color encoding used to highlight differences
- Blue tones indicate higher survival
- Red indicates lower survival

DESIGN CHOICES:

- Minimalistic layout for clarity
- Consistent color scheme
- Clear labeling and titles

TASKS AND APPROACH

TASKS PERFORMED:

- Data cleaning and preprocessing
- Feature creation
- Visualization development
- Dashboard integration
- Adding interactivity through filters

APPROACH:

A systematic approach was followed starting from data preparation to visualization and finally dashboard creation.

KEY INSIGHTS AND CONCLUSION

KEY INSIGHTS

- Women had significantly higher survival rates
- Higher-class passengers had better survival chances
- Younger passengers were more likely to survive
- Small families had optimal survival probability

CONCLUSION

This project demonstrates how data visualization can reveal hidden patterns in data. The interactive dashboard allows users to explore survival trends effectively and supports data-driven insights.